
Synthetic Cultural Agents from Aggregate Anchors

A Falsifiable Protocol for Constructing Aggregate-Anchored Choice Policies

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Status of this version. This is a methods-first working-paper draft. Its conceptual protocol, proposed rejection design, and scope boundary are stated as commitments. Section 7 reports results from the frozen sixteen-country evaluation (both adapter banks, WVS Wave-7 surface), every headline number re-derived from committed artifacts (V1 ledger, 2026-08-20); the honest negative — prompting beats the adapters on distributional shape while the weights carry the GPS direction — is the centerpiece, and the permuted-anchor placebo and operator-sensitivity arms remain predeclared and pending. This draft is not a public preregistration. A verified USA/Mexico pilot is retained only as an Appendix diagnostic.

Abstract

Social scientists increasingly use language models to simulate survey respondents or to align models with named communities. Both approaches obscure a central question: what exactly enters the model, and what evidence would justify interpreting its output as population-relevant? We propose a constructive alternative. A researcher declares an *aggregate anchor*—a vector of population-level measurements—and freezes a protocol that maps the anchor into synthetic pairwise data and then into a language-model choice policy. Country or identity labels are absent from primary inference prompts. The resulting object is therefore not an imitation of an identified person or group, nor an estimated population response distribution; it is an aggregate-anchor-induced synthetic choice policy. Its scientific value is empirical, not assumed. We specify five testable links: anchor transmission, policy encoding, anchor relevance relative to permuted-anchor placebos, external transport to independent human survey distributions, and sensitivity to upstream generation choices. We implement the protocol with country-level Global Preference Survey anchors, synthetic preference pairs, and Direct Preference Optimization, and report a sixteen-country World Values Survey evaluation against base-model, cultural-prompt, uniform-noise, and human baselines. The results dissociate the two routes into the model: prompting recovers distributional shape, while only the trained weights recover the GPS preference direction and its economic gradient; a permuted-anchor placebo and operator-sensitivity arms remain pending. The paper contributes a transparent way to construct and potentially reject synthetic cultural policies while drawing a sharp boundary around claims it cannot support: individual representation, full preference-distribution recovery, and downstream causal or structural inference.

Keywords: synthetic cultural agents; aggregate anchors; language models; preference optimization; survey simulation; falsification; culture

1. A construction problem, not an impersonation problem

Language models are now routinely asked to stand in for people: a prompt supplies a demographic profile, a country name, or a social identity, and the model is asked to answer a survey “as” that person or population. This practice is attractive because it is cheap and scalable. It is also scientifically precarious. Prompted synthetic respondents can be highly sensitive to wording and model version, and their distributions can differ materially from human survey distributions [1, 3, 9]. More fundamentally, an inference-time persona makes it difficult to distinguish information supplied by the researcher from stereotypes or associations retrieved by the model.

The alternative is not to claim that a model has discovered a culture. It is to construct a model object from declared population-level information and then submit that construction to tests that can fail. We call the resulting object an *aggregate-anchor-induced synthetic choice policy*. The policy is built from a country-level preference profile, but it is not asserted to represent any individual in that country. This boundary matters: evidence that models can flatten or misportray identity groups is a warning against treating synthetic outputs as voices of the people they purport to emulate [24].

Our contribution is a protocol, not a theorem of human-preference recovery. A protocol specifies: which aggregate measurements enter; how they are rendered; how scenarios, response pairs, labels, and quality filters are generated; how the policy is estimated; and which internal, placebo, external, and sensitivity tests decide whether the construction is useful. The

protocol is constructive because it creates a policy under researcher-authored rules. It is falsifiable because each link in the construction makes a separate empirical commitment.

The first implementation uses the Global Preferences Survey (GPS), which reports country-level profiles for trust, risk taking, patience, altruism, and positive and negative reciprocity [13]. These anchors yield synthetic preference pairs; Direct Preference Optimization (DPO) fits lightweight country-specific adapters relative to a common base policy [19]. The planned primary external criterion is a theory-mapped battery of World Values Survey (WVS) item distributions. Crucially, country information is excluded from the primary evaluation prompt. This removes a direct country-token channel; it does not prove that all country-associated information is absent from base-model knowledge, scenario wording, or the profile. Thus the relevant question is not whether a model can retrieve a country label at inference. It is whether a policy induced by a declared aggregate anchor has externally useful behavior under a transparent, fixed procedure.

This paper makes four contributions: a formal statement of the aggregate-anchor construction problem and its scientific object; a modular protocol with anti-leakage, provenance, and reproducibility requirements; an empirically testable design built on five links—anchor transmission, policy encoding, anchor relevance, external transport, and operator robustness; and a predeclared multi-country comparison of the constructed policy against base-model, country-prompt, non-LLM, and permuted-anchor baselines, whose results are deliberately not pre-judged here.

2. Positioning: where this protocol differs

The closest literatures differ in both their information source and the location at which that information enters the model. Early “silicon sample” work conditions models on detailed respondent profiles and asks whether generated responses resemble human samples [1, 2, 6]. Subsequent evidence finds instability, distributional mismatch, and inferential risks in such replacement exercises [3, 4, 9]. A second family fine-tunes models on human preference or survey-response data [22, 16, 5]. These approaches may be appropriate when individual records or empirical response distributions are available; their claim is correspondingly closer to learned human response behavior. A third family evaluates or aligns models against named cultural communities [23].

Table 1: Four approaches to LLM behavior associated with social-science targets. The final row is this paper’s object.

Approach	Training information	Conditioning locus	Central claim
Persona prompting	identity or demographic profile	inference prompt	model imitates a named respondent/group
Microdata fine-tuning	individual responses or distributions	weights and/or prompt	model learns observed response behavior
Cultural alignment	named community preferences or norms	weights and/or prompt	model aligns to an identified target
Aggregate-anchor SCA	synthetic pairs induced by declared aggregate measurements	weights; no country prompt in primary evaluation	a protocol-induced policy has passed stated tests of transport and robustness

The protocol uses familiar computational components but makes no novelty claim for them. A language model provides conditional probabilities over candidate responses [17]; DPO provides a tractable way to fit a policy from pairwise preferences; low-rank adaptation makes separate adapters practical [19, 14, 12]. Our contribution is to specify how aggregate measurements enter the synthetic-data operator and how the resulting policy is subjected to tests that distinguish construction success from merely plausible prose.

3. The scientific object and its boundary

3.1 Aggregate anchors

For country c , let

$$z_c = (z_{c1}, \dots, z_{cK}) \in \mathbb{R}^K \quad (1)$$

be a vector of pre-existing population-level measurements selected before training. In the first implementation, $K = 6$ and z_c contains GPS country scores, represented in the pipeline as standardized deviations from global means. An aggregate anchor is an input to a construction procedure. It is *not* an individual record, a country name shown at inference, or a moment

condition automatically satisfied by a trained policy.

We call such an input a *declared* anchor: it is chosen by the researcher from a pre-existing, citable measurement source under a written inclusion rule, and it is fixed before training and before any outcome is inspected. A declared anchor is thus a researcher-committed input, whereas an *identified* anchor is one recovered from the data.

3.2 A frozen construction protocol

We define a protocol as

$$\mathcal{P} = (Z, R, G, S, Q, A, E), \quad (2)$$

where Z records the anchor source and country frame; R is the deterministic anchor-to-profile rendering rule; G is the scenario, response-pair, and label-generation procedure; S is scoring, quality control, and filtering; Q records prompts and direct-token anti-leakage restrictions; A is the policy estimator and its training configuration; and E is the planned evaluation, placebo, and sensitivity design. Conditional on an archived run manifest, the model-mediated stages of G and S are stochastic realizations; the protocol is not equivalent to a deterministic mapping from z_c to data. In the implemented protocol, Z is the GPS ingestion and country frame of Section 6.1, R the deterministic profile renderer, and G the teacher–generator bank together with deterministic sign labeling (Section 4.2). S is inherited pair scoring used as a diagnostic, Q the frozen prompts, A the DPO fit of Section 4.3, and E the tests of Section 5.

Given a reference policy π_{ref} , the protocol generates a country-specific synthetic dataset $\mathcal{D}_c = \mathcal{GP}(z_c)$ and fits

$$\hat{\pi}_c^{\mathcal{P}} = \mathcal{A}(\mathcal{D}_c; \pi_{\text{ref}}). \quad (3)$$

Equation (3) defines a conditional distribution over candidate responses induced by a declared anchor and a frozen protocol. It does *not* assert $\hat{\pi}_c^{\mathcal{P}} = P_{\text{human},c}$, where $P_{\text{human},c}$ denotes a human response distribution. That equality is an empirical transport hypothesis, not a definitional fact.

3.3 What this construction does not identify

Country averages do not identify which individuals hold which preferences, how preferences co-vary with demographic characteristics, or the full within-country distribution. The protocol therefore does not claim to recover individual heterogeneity, subgroup distributions, or a country’s latent culture. Nor does it license reduced-form causal estimation, structural parameter recovery, or policy counterfactuals without a separate measurement and identification argument. These are not rhetorical caveats; they determine the object of study.

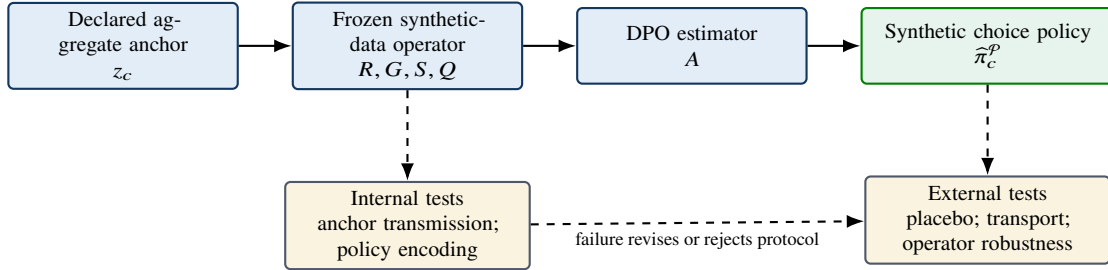


Figure 1: The paper’s scientific object. Solid arrows construct a policy. Dashed arrows are tests, not guarantees. A failed test rejects a claim about this protocol; it does not prove that a population lacks the relevant preference.

4. From anchors to synthetic choice policies

4.1 Anti-leakage and profile rendering

The primary implementation begins with the GPS vector and renders it into an anonymized quantitative profile. The rendering states the magnitude and direction of each standardized score without inserting a country name. In the primary evaluation prompt, country and demographic labels are likewise omitted. This direct-token anti-leakage design does not establish cultural validity or prove that country-associated information is absent. It establishes a narrower, auditable fact: any difference between adapters cannot be attributed to country tokens in the primary inference prompt. The claim must be complemented by audits of scenario text, profile re-identifiability, and whether a classifier can infer country from generated artifacts.

4.2 Synthetic data operator

The operator has two layers. A teacher first proposes behaviorally distinct facets and culturally neutral scenarios for each preference dimension, and a generator then writes two opposing responses to each scenario while holding non-target traits

constant. Response A is instructed to load positively on the target dimension and Response B negatively. That shared bank does not depend on the country. Country identity enters only as a label: the production rule assigns the high-loading response when $z_{c,k} \geq 0$ and the low-loading response when $z_{c,k} < 0$. An independent scorer may assign both responses six-dimension scores; in the implemented protocol those scores are inherited from the materialized bank and used as diagnostics. Contamination is recorded and is not a retention gate. A legacy LLM selector that was prompted to implement the same sign rule is retained only as a sensitivity arm.

Example (illustrative). Mexico’s GPS patience score is -0.11 . The rendering stage states that standardized patience is moderately below the global mean. The teacher proposes a culturally neutral intertemporal scenario, and the generator fixes two opposing responses: A waits for the larger later reward (positive loading) and B takes the smaller reward now (negative loading), with risk attitudes and income constraints held as constant as the prompt can make them. The sign rule, seeing a negative z , labels B as chosen for Mexico. Inherited scorer ratings, when available, are attached as diagnostics and do not decide admission. This trace illustrates the implemented mechanics; it is not an empirical result.

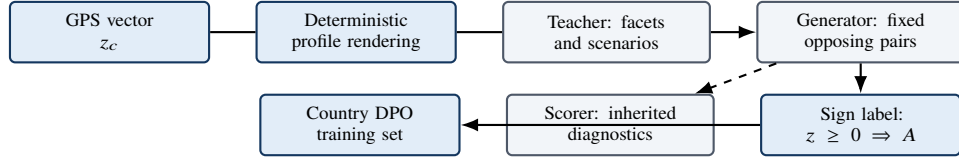


Figure 2: Implemented architecture. Profile rendering and sign labeling are deterministic; the teacher and pair generator are the stochastic bank. The scorer is diagnostic and is not a contamination gate. An LLM selector is a sensitivity arm, not this path.

This decomposition matters. Calling every upstream choice “teacher bias” obscures the actual sources of dependence: scenario construction, response-pair generation, the labeling rule, scorer judgments, and any filter thresholds may each change the synthetic training distribution. The empirical design therefore treats the full operator as an object of sensitivity analysis.

4.3 Policy estimation

For synthetic triplets $(x, y_w, y_\ell) \in \mathcal{D}_c$, DPO fits a policy relative to π_{ref} by minimizing

$$\mathcal{L}_{\text{DPO}}(\pi) = -\mathbb{E}_{(x, y_w, y_\ell) \sim \mathcal{D}_c} \left[\log \sigma \left(\beta \log \frac{\pi(y_w | x)}{\pi_{\text{ref}}(y_w | x)} - \beta \log \frac{\pi(y_\ell | x)}{\pi_{\text{ref}}(y_\ell | x)} \right) \right]. \quad (4)$$

DPO is a pairwise-preference estimator with reference-policy regularization [19]. Here it fits synthetic preferences generated by $\mathcal{P}$. It is not a constrained estimator that mechanically enforces $\mathbb{E}[f(Y)] = z_c$, and we make no such claim. Intensity-weighted pairwise variants that down-weight near-zero anchors are natural sensitivity arms in E [20].

5. Making the construction empirically rejectable

The phrase “falsifiable” is useful only if it names propositions, estimands, and failure consequences [11, 8]. This working-paper version supplies the test architecture, but not a public timestamped registration or final numerical thresholds; it should therefore be read as a proposed rejection design. The archived analysis plan fixed the unit of analysis, primary metric, comparison, practical margin, randomization/materialization count, uncertainty procedure, and country/domain/protocol decision rule for every row in Table 2 before outcomes were inspected. The five links have now been exercised by the sixteen-country evaluation, with per-link status in Table 2. Each test concerns a distinct link; passing one does not imply passing another.

Table 2: The five rejection points of an aggregate-anchor construction.

Claim	Predeclared evidence	evi-	Status after the sixteen-country evaluation	Interpretation of failure
Anchor mission	trans-label/sign agreement, separation, rates, anchor swaps	agree-target QC	PASSED — deterministic sign labeling ($G = \text{sign}(z)$) is the default; label/sign agreement verified on the production 658-pair bank	the synthetic operator does not express the declared input reliably
Policy encoding	held-out pair recovery; policy-distance stability	synthetic-recovery;	PASSED — held-out synthetic-pair accuracy 0.886–0.992 ($n = 132$ per country)	DPO does not retain the contrast present in the synthetic data
Anchor relevance	real-anchor versus permuted-anchor placebo adapters	versus	PENDING — restricted permutation placebo not yet run; required before claiming anchor information beyond generic synthetic tuning	apparent gains may reflect generic synthetic tuning rather than anchor information
External port	trans-independent human survey distributions versus base, prompt, and non-LLM baselines		PARTITIONED — distributional shape: adapters (TVD 0.469) lose to the persona prompt (0.375) and uniform noise (0.338); construct bridge: adapter trust $\rho = 0.70$ vs. persona 0.06 (human layer 0.41), development partials +0.56 (trust) / +0.46 (patience). Verdict: no histogram reproduction; preference direction and its economic gradient transport	this protocol does not generalize to the chosen external criterion
Operator robustness	scenario, generator, selector, scorer/QC, materialized-data, and training-seed ablations		PARTIAL — inference-time temperature scaling measured (pooled TVD 0.469 $\rightarrow$ 0.332 at $T = 3$, top-option match unchanged); training-side ablations pending (extended-prompt bank, soft-DPO/ $ z $ pruning, repeated seeds, deterministic-vs-legacy selector)	the conclusion depends on arbitrary or unstable upstream choices

5.1 Internal tests: transmission and encoding

Anchor transmission asks whether changing z_c changes labels and retained pairs in the direction dictated by the declared rule. It is assessed with country-by-dimension label summaries, quality-control pass rates, target-versus-nontarget separation, contamination ratios, and anchor-swap diagnostics. The current implementation makes an especially important test available: because the selector is instructed to implement a sign rule, the primary sensitivity comparison must contrast a deterministic sign-labeling operator with the legacy LLM selector. Policy encoding asks whether the trained adapter reproduces held-out synthetic contrasts and whether independently trained policies are stable under repeated materializations of the same protocol.

5.2 Anchor relevance: a restricted permutation distribution

A central negative control draws many restricted permutations of country anchors before synthetic generation while retaining all other stages. The final plan will predeclare any strata used to preserve region, sample availability, or anchor magnitude, and will evaluate the real-anchor statistic in its resulting empirical null distribution. Each placebo uses the same data volume, model family, and training budget as the reference protocol. If real-anchor adapters do not outperform this distribution on predeclared external outcomes, the data do not support the claim that aggregate anchors add useful information beyond generic synthetic fine-tuning.

5.3 External transport and uncertainty

External transport is a predictive-criterion claim: the policy’s induced option probabilities improve alignment with independently collected human response distributions. It is neither recovery nor validation of a complete country distribution. For a question q with valid options $\mathcal{Y}_q$, let $\ell_{\hat{\pi}_c}(y | q)$ denote the total log probability assigned to the token sequence encoding answer option y under a fixed prompt and option-order rule. The policy yields

$$\hat{P}_c(y | q) = \frac{\exp\{\ell_{\hat{\pi}_c}(y | q)\}}{\sum_{j \in \mathcal{Y}_q} \exp\{\ell_{\hat{\pi}_c}(j | q)\}}, \quad (5)$$

and the primary descriptive discrepancy is total variation distance (with frequency-based fixation indices [18] as a dispersion-sensitive alternative),

$$\text{TVD}(\hat{P}_c, P_c) = \frac{1}{2} \sum_{y \in \mathcal{Y}_q} |\hat{P}_c(y | q) - P_c(y | q)|. \quad (6)$$

For each country–item cell, P_c is the survey-weighted empirical option distribution after a frozen harmonization map for option wording, order, invalid outputs, and Likert categories. The analysis will report absolute TVD and paired differences from every baseline, together with practical margins. It will distinguish finite-survey sampling uncertainty, country and item aggregation, materialized synthetic datasets, and DPO training seeds; a hierarchical or randomization-based aggregation rule will be fixed before outcome inspection. A bootstrap over survey items is useful only as a robustness calculation for the declared item battery—not as respondent sampling uncertainty, an item-universe inference, or uncertainty about an upstream model-mediated pipeline.

6. Empirical implementation: predeclared design

6.1 Data roles and country frame

The first implementation uses GPS as the anchor source and wvs Wave 7 as the primary independent outcome surface. Before a scaled run, the authors will archive an immutable country inclusion/exclusion rule, a named 10–15-country panel from the GPS×wvs intersection, and a frozen theory-led item battery. The battery must separately list construct-linked confirmatory outcomes—with an expected direction or distributional relation for each GPS dimension—and descriptive/exploratory outcomes. The existing 35-item pilot map contains 30 mapped items and five demographics; it is not itself the confirmatory map for the scaled study.

AmericasBarometer and the Integrated Values Surveys may serve as secondary instrument or temporal replications if multi-country overlap is adequate. World Bank indicators are contextual covariates for exploratory heterogeneity analysis, not direct choice-distribution criteria.

6.2 Anchor sources: criteria, GPS, and alternatives

A source must satisfy a small set of conditions before it can anchor the construction. It must be pre-existing and citable, fixing the reference in the public record rather than constructing it for this study; it must offer population-level aggregates rather than individual microdata, because the outcome surface is itself country-level; and it must be theoretically linked to

the constructs of interest while remaining independent of the evaluation surface, so that anchoring on wvs country means while evaluating wvs items would be circular. It must also offer broad country coverage and valid, comparable measurement across countries.

GPS is a defensible first implementation of these criteria. Its country scores are aggregates by construction, derived from measures of six economic preferences that were validated *ex ante* in incentivized experiments, collected in representative samples across 76 countries—a set that overlaps the wvs Wave 7 country frame. The data are public and citable, and the criteria are fixed before any scaled run.

Alternatives exist if the criteria recommend them: wvs/EVS country means, with the caveat of overlap with typical evaluation surfaces; Schwartz value country scores (PVQ/ESS); Hofstede dimensions; Inglehart–Welzel cultural-map coordinates [15]; and the GLOBE project. Table 3 lists these candidates with their principal caveats. The method is agnostic to the specific source, so long as the criteria above hold.

Table 3: Candidate anchor sources and their principal caveats.

Source	What it measures	Key caveat
GPS	Six economic preferences (trust, risk-taking, patience, altruism, positive and negative reciprocity)	Aggregate by construction; experimentally validated
wvs/EVS means	Country-level value orientations and attitudes	Circular when the evaluation surface is wvs itself
Schwartz (PVQ/ESS)	Ten basic human values at country level	Fewer countries; values are not preference measures
Hofstede dimensions	Work-related cultural dimensions	Dated samples; dimensions are not preference measures
Inglehart–Welzel map	Emancipative and secular value coordinates	Two-dimensional reduction of a larger item space
GLOBE	Societal practices and leadership values	Organizational focus; limited item overlap

6.3 Baselines and primary comparison

Every country is evaluated against four mandatory baselines: (i) the unmodified base model; (ii) a country-named cultural-prompt baseline; (iii) a named non-LLM aggregate benchmark fixed before outcome inspection; and (iv) the restricted permuted-anchor distribution. The country prompt is deliberately included despite violating direct-token anti-leakage because it answers a distinct practical question: whether an explicit persona retrieves useful information more effectively than the constructed policy. The analysis must report provenance-sensitive construction success separately from predictive alignment; a cultural prompt may be practically predictive while failing the former objective.

The primary cross-country claim will compare each fixed policy to each country’s human distribution. Under unconditioned prompts, this is *fixed-policy proximity*, not evidence that the model conditions on a country at inference. Country-level permutation tests and cross-country patterns become informative only with more than two adapters. The design will report gains and failures, rather than defining success as a uniformly positive average.

6.4 Operator-sensitivity panel

A stratified 4–6-country subset is run before the full panel. The reference protocol is compared with alternative scenario banks, alternate pair-generation materializations, deterministic-sign versus legacy-selector labels, QC-threshold variations, a scorer-diagnostic-only/no-QC arm where feasible, restricted anchor permutations, and repeated DPO seeds. Each arm reports effects on (i) synthetic-data diagnostics, (ii) learned-policy stability, and (iii) external transport. At least three independently materialized scenario/pair banks are retained per condition. The full panel proceeds only if this subset establishes adequate anchor transmission, excludes obvious high-contamination pairs under a predeclared rule, and rejects the basic placebo explanation.

7. Results: sixteen countries, four surfaces

All numbers in this section were re-derived from the frozen evaluation artifacts by the committed pipeline `analysis/phase2/` (V1 verification ledger, 2026-08-20); none are quoted from secondary summaries. Sixteen country-specific adapters — CHN, JPN, GBR, USA, MEX, ARG, DEU, RUS (first bank) and IND, IDN, NGA, EGY, TUR, NLD, BRA, GRC (second bank) — were trained on 526 synthetic sign-labeled pairs drawn from the shared 658-pair bank ($G = \text{sign}(z)$, $\beta = 0.1$, LoRA $r = 16/\alpha = 32$, one epoch; Appendix C) and evaluated on the wvs Wave-7 surface: 30 single-choice items (binary and ordered scales) mapped to the six GPS dimensions through the theory-led map. The single-choice surface follows the evaluation notebook’s construction: the mapped items less the multi-select childrearing battery (Q12–Q13–Q14), plus the three single-choice demographic items (Q260, Q275, Q288); the multi-select and numeric-open items are excluded because their option space is not a single softmax. Four model families are compared on identical surfaces: the adapters, the unconditioned base, the base prompted as a resident of each country (persona), and uniform noise; the survey-weighted human distributions are the target.

7.1 Distributional fit: prompting wins, but the population is diffuse

Table 4 reports the pooled distributional surface (means over item×country cells).

Table 4: Distributional surface, pooled across 16 countries and 30 single-choice items. Lower TVD/JSD and smaller $|\text{entropy err}|/|\text{std err}|$ are better; top-match is a rate. Base is one fixed distribution replicated across countries.

Model	TVD	JSD	Entropy err	Std err	Top-match
adapter	0.469	0.279	−0.737	−0.647	0.452
base	0.443	0.249	−0.540	−0.535	0.452
persona	0.375	0.188	−0.438	−0.449	0.479
noise	0.338	0.145	+0.568	+0.845	0.331

The first result is a floor, not a model. Uniform noise achieves TVD 0.338 — *below* every model — because the wvs populations are diffuse: on many items the population is near-uniform, so any concentrated distribution pays a TVD penalty and TVD rewards flatness as much as accuracy. Against this floor, the persona prompt is the best model family: pooled TVD 0.375, only 0.037 above noise, with the smallest under-dispersion (entropy error −0.44) and the best top-option match (0.48). The persona beats the adapters in 15 of 16 countries and the base in 15 of 16.

Why do the adapters underperform on TVD? They are the most over-concentrated family (entropy error −0.74, standard-deviation error −0.65), and they beat the base in only 5 of 16 countries. This is a direct consequence of hard pairwise DPO labels: a pairwise preference signal identifies an ordering, never population marginals. It replicates the mode-collapse diagnosis of Pfeifer & Dalloul (2026) on the same base model and of the wider synthetic-survey literature. Distributional shape is precisely what the weights-only construction does not deliver.

7.2 The construct bridge: direction lives in the weights

Table 5 reports Spearman correlations between country composites and GPS country z -scores, per dimension and family.

Table 5: Construct bridge: Spearman ρ of the country composite vs the GPS country z -score. Human layer $n = 42$; adapter and persona layers $n = 16$. Base and noise are single fixed distributions with no cross-country variation (undefined by construction).

Dimension	Human (42)	Adapter (16)	Persona (16)
patience	0.32	0.42	0.60
risktaking	−0.08	0.57	−0.11
posrecip	−0.00	0.55	0.02
negrecip	0.42	0.06	0.34
altruism	0.03	0.24	−0.23
trust	0.41	0.70	0.06

The key contrast is on trust, the strongest GPS proxy in this menu (Falk et al. 2018’s published country-level correlation is 0.49). The adapter composite ranks countries in GPS order ($\rho = 0.70$)¹; the human wvs layer does too (0.41); the persona prompt does *not* (0.06). On patience the dissociation is dimension-specific: the adapter tracks (0.42) and so does the persona (0.60), against a human layer of 0.32 — the persona’s patience ordering is plausibly inherited from development-correlated priors (Section 7.3). The remaining dimensions show the *imprint* pattern: risktaking (0.57), positive reciprocity (0.55), and altruism (0.24) show adapter correlations far *above* the human layer’s (−0.08, −0.00, 0.03). Because those wvs items do not track GPS in humans, a high adapter ρ there is the GPS sign bit expressed through a poorly mapped instrument — an imprint, not a recovered survey construct, and a reminder that cross-national item comparability is the precondition for construct interpretation [7, 21, 10]. Figure 3 displays the bridge across all six dimensions, with base and noise undefined by construction.

¹Composites use the unified pipeline’s exact-equality recode of binary items, with missing codes masked before recoding. Under the alternative majority-threshold recode the adapter trust bridge is 0.79; the unified construction is reported throughout, and the difference is a construction sensitivity rather than a fixed constant.

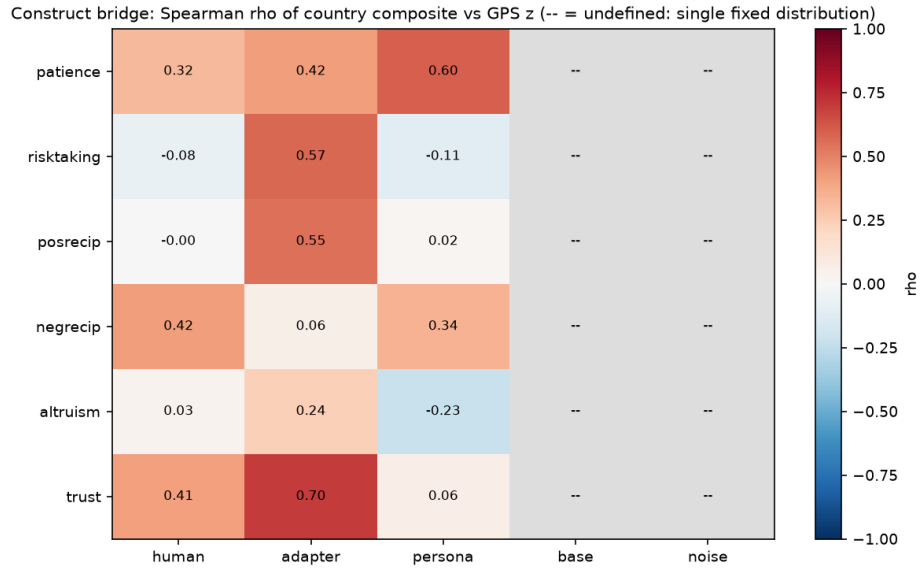


Figure 3: Construct bridge: Spearman ρ of the country composite versus the GPS country z -score, by dimension and model family. Gray cells are undefined (base and noise are single fixed distributions with no cross-country composite variation). The trust row is the paper’s positive result: adapter 0.70, human layer 0.41, persona prompt 0.06. Human layer $n = 42$; adapter and persona layers $n = 16$.

7.3 Development restrictions: the adapters carry the anchor’s gradient

Table 6 reports Spearman correlations of each composite with log GDP per capita (WDI 2015–2019), with education partials.

Table 6: Development restrictions: Spearman ρ of the country composite vs log GDP per capita (WDI 2015–2019); the adapter column also reports the partial controlling education (WVS Q275). Base and noise undefined by construction.

Dimension	Human (42)	Adapter (16)	Adapter partial edu	Persona (16)
patience	0.09	0.29	0.46	0.33
risktaking	−0.34	0.35	0.33	−0.62
posrecip	−0.22	0.03	0.46	0.63
negrecip	0.42	−0.32	−0.49	−0.59
altruism	−0.41	0.27	0.54	0.50
trust	0.09	0.13	0.56	0.73

The adapters reproduce the GPS development gradient on the confirmatory dimensions: trust +0.13 (education partial +0.56) and patience +0.29 (+0.46), in the same direction as the GPS anchor itself (patience +0.58, trust +0.32 against log GDP per capita over the 76-country bank; 75 countries have WDI GDP coverage) and consistent with the human wvs layer (trust +0.09). The persona *overshoots* (trust +0.73), consistent with prompting inheriting the base model’s Western-coded priors. The base and noise, being single fixed distributions, carry no country variation and are undefined on this surface by construction.

7.4 Encoding and direction

The internal encoding link passes: held-out synthetic-pair accuracy is 0.886–0.992 across the sixteen countries ($n = 132$ per country; 0.89–0.99 rounded), so the adapters retain the contrasts present in their training data. Table 7 reports top-option match by dimension. The adapters’ modal answers match the population mode on positive reciprocity (0.78) and trust (0.52). The noise baseline reaches 1.00 on altruism and 0.77 on negative reciprocity purely because those populations are near-unimodal — top-option match alone is not a validity statistic.

Table 7: Top-option match rate by dimension and model. Noise reaches 1.00 on altruism and 0.77 on negative reciprocity only because those populations are near-unimodal.

Dimension	Adapter	Base	Persona	Noise
patience	0.16	0.03	0.03	0.12
risktaking	0.28	0.27	0.08	0.48
posrecip	0.78	0.87	0.87	0.10
negrecip	0.50	0.62	0.64	0.77
altruism	0.54	0.33	0.81	1.00
trust	0.52	0.52	0.57	0.15

7.5 Cultural distance: two geometries of (non-)specificity

Cultural distance is not a single quantity, and the two geometries we compute disagree in an instructive way. On the frequency-based F_{ST} geometry of Muthukrishna et al. (2020) — which treats questions as loci and response distributions as alleles — the median rank of an adapter’s own country among the 42 WVS populations is 20.5, indistinguishable from chance (21.5); 11 of 16 adapters are nearest to Canada, a mid-scale reference, and the best adapters (GBR rank 2, RUS rank 4, BRA rank 8) sit far from their own countries. On the location geometry — the pooled TVD of each adapter’s fixed output distribution against all 42 populations — the median own-country rank is 15.0 (chance 21.5): weakly but systematically below chance, with GBR, GRC and USA each at rank 2, and no adapter closest to its own country (Supplementary Figure S7). The two metrics thus disagree about how much location information survives, and the disagreement is informative rather than contradictory: F_{ST} is a variance-ratio that penalizes the adapters’ over-concentration, while TVD reads the location shift the anchors induced. The PCA of the mapped items tells the same story as the frequency geometry: adapter and persona positions track the model family, not the target country (Figure 4, right).

The nearest-country structure is dominated by a mid-scale attractor: Canada is the nearest of the 42 populations for 11 of 16 adapters on both geometries. This is not evidence of a Canadian cultural affinity in the adapters — Canada sits near the cross-country center of the item space, so it is the closest point for any distribution whose location signal is weak. The adapters whose own-country fit is worst are precisely the sign-twin / development-shadow set from the construct analysis: NGA (rank 28), IDN (31), MEX (31), EGY (35) and IND (40) all fall below chance, and their GPS profiles are nearly collinear with GDP.

The coarse structure that does survive is the sign profile. The 76-country GPS bank collapses to 35 sign(z) profiles (22 multi-country classes), and the sixteen adapters instantiate 13 distinct profiles (MEX $\equiv$ RUS, IND $\equiv$ GRC, IDN $\equiv$ EGY). Sign-twin proximity on human wvs is explained by shared GDP (partial correlation with the GDP gap -0.09): country-level sign patterns are a development shadow, not cultural families, and the adapters inherit this structure by construction because the anchor enters through its sign.

What, then, do these objects carry of the countries that anchor them? The imprint pattern of the construct-bridge analysis is the answer in miniature. On dimensions where the wvs instrument tracks GPS in humans (trust, patience), the adapters’ country ordering aligns with the anchor ($\rho = 0.70$ on trust) and the development gradient survives in sign; on dimensions where the instrument does not (risktaking, positive reciprocity, altruism), the adapters still express the anchor’s sign bit, but through a poorly mapped instrument — an imprint, not recovered culture. The adapters are therefore identified for the anchor’s aggregate preference ordering, not for country-specific dispersion, magnitudes, or latent cultural structure. Prompting buys the complementary half of the dissociation — distributional shape without ordering. Neither route reproduces a country’s culture; each carries a provable fragment of it, and this section has measured which fragment.

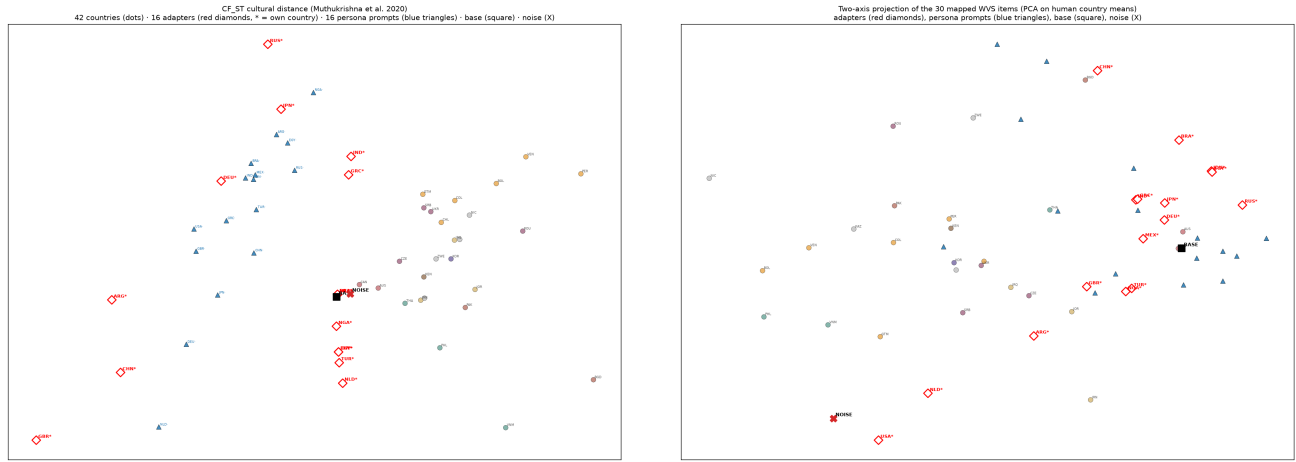


Figure 4: CF_ST cultural distance (left; Muthukrishna et al. 2020) and two-axis PCA of the mapped items (right). Red diamonds are adapters (* = own country), blue triangles persona prompts, the square the base, × noise; country dots colored by region. Adapters do not localize near their own countries on either geometry.

7.6 The dissociation

The results dissociate the two routes into the model. The country-named prompt buys distributional shape (TVD 0.375, best dispersion and top-match) but carries no GPS ordering on trust ($\rho = 0.06$) and overshoots the development gradient (+0.73). The trained weights carry the GPS direction — trust bridge $\rho = 0.70$, development partials +0.56/+0.46, held-out pair accuracy up to 0.99 — but pay for it in concentration and in TVD. The anti-leakage price is ≈ 0.09 TVD: the adapters’ pooled TVD (0.469) exceeds the persona’s (0.375) by 0.094. What we cannot claim follows: histogram reproduction, country-specific dispersion, or magnitudes. These adapters are identified for preference *direction* with transparent provenance, not for survey-distribution point prediction.

7.7 Operator diagnostics and the dispersion bound

Temperature scaling is an inference-time diagnostic. Raising the sampling temperature to $T = 3$ moves pooled TVD from 0.469 to 0.332 while the top-option match is unchanged (0.452): it recovers dispersion but does not move the location of the induced distribution. The shape loss is therefore a softmax artifact of low-temperature sampling, whereas the location loss is structural. The intensity-weighted training analogue of this lever is studied by [20].

The training-side levers were predeclared in Table 2. An extended-prompt training bank — richer completions, not binary A/B — is the only training-side lever that can change the shape of the learned distribution without touching wvs. Soft DPO or $|z|$ -pruning of near-zero anchors is a cheap ablation for knife-edge dimensions (patience, risktaking); we expect it to affect direction, not dispersion.

An explicit anti-contamination commitment: we will not train on population marginals. KL-to-wvs or moment-matching losses would contaminate the out-of-sample design, in which wvs stays unseen. Temperature scaling is reported as a diagnostic, not as a training change.

8. Discussion

The protocol changes the question posed to synthetic cultural agents. It does not ask whether an LLM can speak for a country, nor does it assume that country-level averages identify individual preferences. It asks whether a declared aggregate profile can be transformed into a reproducible policy whose synthetic provenance is visible and whose external relevance is testable.

This distinction has practical consequences. The method may be useful for exploratory measurement, hypothesis generation, survey-item design, and benchmark construction when its tests support those applications. It is not a substitute for collecting human data, and it should not be used to infer individual beliefs, to represent subgroups, or to support causal and structural claims without additional evidence. The correct scientific outcome is sometimes rejection: an anchor may fail to transmit, an adapter may fail to encode, a placebo may perform similarly, or an external criterion may not transport.

The protocol also makes clear why operator sensitivity is substantive. If a finding disappears when a scenario bank, selector, quality-control threshold, or training materialization changes, then the finding belongs to a fragile synthetic pipeline rather than to the aggregate anchor. Conversely, stability across those choices does not prove representation, but it increases confidence that the observed policy behavior is an attribute of the specified construction rather than an incidental artifact.

Code and data availability

The SCA2 implementation, frozen evaluation artifacts, and this working-paper source are available at https://github.com/EconLLM-Lab/SCA2_PoFW. A claim-ready release will identify an immutable repository tag/commit, code license, model/API versions, and an artifact manifest. GPS data are documented by Falk et al. [13]; wvs Wave 7 data and documentation are available from the World Values Survey Association under DOI <https://doi.org/10.14281/18241.18> [25]. Raw survey microdata are not redistributed with the repository.

A. The two-country pilot: a fixed-policy proximity diagnostic

This pilot predates the sixteen-country evaluation and is retained as a design-boundary illustration, not as evidence: it is superseded by the main results (Section 7). The existing two-country pilot is retained because it clarifies a design boundary. Under the primary unconditioned prompt, the same model emits identical option probabilities when scored against USA and Mexico; only the external target distribution differs. Thus a matched-versus-cross comparison measures which adapter’s *fixed* output distribution lies closer to each population’s observed responses. It does not show that the adapter knows or conditions on its target country at inference.

In the committed pilot, the Mexico adapter has mean TVD 0.4882 on Mexico versus 0.5211 for the base model, while the USA adapter has 0.6377 on Mexico. The corresponding mean matched-minus-cross TVD difference is 0.1495 with a question-resampling 95% interval [0.0964, 0.2152] across 35 items. On USA outcomes, however, the Mexico adapter is approximately neutral relative to base (0.3888 versus 0.3958), while the USA adapter degrades (0.5258). These values are verified transcriptions of the committed evaluation summaries.

The pilot therefore does not establish country specificity. A single fixed adapter can dominate another fixed adapter against either country, and $C = 2$ cannot reveal a stable country-by-policy cross-over pattern. The item bootstrap quantifies sensitivity to this 35-item battery, not uncertainty from training runs, model generation, respondents, or country sampling. The pilot motivates—rather than substitutes for—the multi-country placebo and operator-sensitivity design in the main paper.

B. Adversarial scientific-design audit of the synthetic-generation pipeline

Scope and evidence. This independent appendix is a static audit of the specified pipeline and its tests, not an empirical validation of generated records. The targeted test suite was attempted but could not be collected in the review environments because `pandas/numpy` dependencies were unavailable. The inspected tests predominantly mock model completions (for example, `synthetic_generation/tests/test_generate.py:139–184` and `test_score.py:13–182`); no test-pass, psychometric-validity, inter-rater-reliability, or model-stability claim follows from this audit.

B.1 Blocker findings

Replace the LLM selector with a deterministic sign rule for non-zero target scores. The selector is explicitly instructed that positive z implies A and negative z implies B, while fixed A/B responses were already generated as positive/negative target loadings (`generate.py:170–180, 237–279`). It is therefore a stochastic, prompt-mediated implementation of a supplied deterministic label rather than an independent estimator of country-specific preference. It introduces label noise, permits irrelevant use of the other five profile dimensions, and depends on mutable model behavior. The primary construction should use A when $z > 0$ and B when $z < 0$, with $|z|$ near zero excluded, flagged, or handled under a separately justified probabilistic rule. The current exact-zero fallback is internally inconsistent: selector guidance asks for the “less extreme” option whereas QC treats zero as positive (`generate.py:250–255; score.py:204–206`).

Treat LLM scoring/QC as process diagnostics, not a validity screen. The scorer receives the target dimension and rubric, scores the already selected chosen/rejected texts, and QC retains rows whose scorer-derived target difference agrees with the known GPS sign (`score.py:46–64, 119–130, 204–235`). This is a second model confirming a target specified by design, not an independent measurement of cultural preferences or construct validity. No calibration against blinded human annotations, reliability analysis, held-out behavior, or threshold justification is present in the inspected code/tests. The scores may be useful operational diagnostics, but should not be described as validating a population-level construct.

B.2 Major findings

Selection-on-the-outcome and unfiltered contamination threaten validity. Scoring follows country-specific selection, so fixed A/B pairs are repeatedly scored in country-oriented order rather than once blind to country and choice (`generate.py:429–474; score.py:119–146`). Cross-dimension movement is calculated but does not determine `qc_status`; export removes only non-pass statuses, so high-contamination pairs can remain (`score.py:209–221; export.py:323–329`). The recommended design scores canonical A/B options once under blinded target and option order, orients scores arithmetically after deterministic labeling, reports all cross-loading distributions, and either predeclares a contamination gate or treats contamination as descriptive only.

Prompts attempt control but do not demonstrate it. The pair generator is instructed to vary only the target trait while holding the other five constant (`generate.py:175–180`); this is an instruction, not evidence of compliance. Optional curated anchors are injected into scenario and triplet prompts (`generate.py:72–76, 156–160`), creating a provenance channel that must be versioned and separated from evaluation materials. Country names are absent from profile text (`profiles.py:44–65`), a useful safeguard, but all six numeric dispositions remain available to selection (`generate.py:263–275`).

Generation reproducibility is incomplete. The manifest records config snapshots, GPS vectors, costs, scenario counts, timestamp, and a short Git hash; export shuffling uses a seed (`export.py:41–50`, 367–399). However, the pipeline seed is not passed to model calls and teacher/generator temperatures are non-zero (`config.py:225–228`; `generate.py:41–49`, 194–202). Manifests lack input hashes, resolved model revisions, dependency locks, raw-checkpoint hashes, prompts, and repository dirty state. Resume can reload current input data before rescoring an old checkpoint (`run.py:363–418`). Outputs should therefore be treated as dated stochastic realizations until immutable input and full-run provenance are archived.

B.3 Efficiency-preserving changes and a minimal ablation

Deterministic signs eliminate all per-country selector calls. The default 120 fixed triplets (six dimensions times 20 scenarios) otherwise generate 120 times the number of countries selection calls. Score each canonical pair once rather than once per country, orient scores locally by country sign, and compute QC arithmetic locally; this reduces country-linear scorer calls and removes country/choice-order leakage. Facets and scenarios are already batched, but per-triplet selector reasoning payloads are not used for labeling or QC and can be omitted or sampled for audit, while preserving raw responses and request metadata for remaining calls.

The minimum ablation uses a fixed scenario/A–B bank and compares: (i) deterministic signs versus legacy LLM selection; (ii) no QC versus scorer diagnostic-only versus scorer plus a predeclared contamination exclusion; and (iii) anchors off versus on. Each condition should retain at least three independently generated banks, stratified by dimension and z -score magnitude. Report label disagreement with the deterministic rule, QC retention, cross-loading distributions, blinded human ratings of target loading/non-target leakage/plausibility/option balance, and the full range of downstream conclusions. If legacy-selector and deterministic-sign estimates differ materially, the former is an unvalidated alternative rather than the primary dataset.

Resolution. Since this audit was conducted, the committed pipeline made deterministic sign labeling the default labeling rule ($z \geq 0 \Rightarrow A$); the legacy LLM selector is retained only as a sensitivity arm, LLM scoring/QC output is treated as process diagnostics rather than a validity screen, and contamination is recorded and is not a retention gate.

C. Training configuration

The reference training configuration in Table 8 is fixed before any scaled run; every adapter in the pilot and scaled protocol is estimated against it, and any deviation from it is declared as a sensitivity arm rather than an ad hoc adjustment. Entries are transcribed from and verified against the canonical training artifacts under `DPO_train_test/`. The production data row refers to the $\text{sign}(z)$ production bank (658 pairs, 526/132 split), which supersedes the legacy teacher-generator bank ($\approx 3,594$ triplets, 2,875 train) recorded in `DPO_train_test/`.

Table 8: Reference training configuration for the country-specific DPO adapters. All variants in the operator-sensitivity panel are declared relative to this configuration.

Component	Setting	Source/status
Base model	Llama-3.1-8B-Instruct (meta-llama/Llama-3.1-8B-Instruct)	DPO_train.ipynb
QLoRA quantization	4-bit NF4, double quant, fp16 compute	DPO_train.ipynb
LoRA rank/alpha	$r = 16$, $\alpha = 32$, dropout 0.05	DPO_train.ipynb
LoRA target modules	<code>q_proj</code> , <code>k_proj</code> , <code>v_proj</code> , <code>o_proj</code> (attention projections)	DPO_train.ipynb
DPO beta	$\beta = 0.1$	DPO_train.ipynb
Learning rate/schedule	1×10^{-4} , cosine, warmup ratio 0.03	DPO_train.ipynb
Epochs	1	DPO_train.ipynb
Batch size/gradient accumulation	per-device batch 1; 16 accumulation steps (effective 16)	DPO_train.ipynb
Synthetic triplets per country	526 of the shared 658-pair $\text{sign}(z)$ bank; 132 held-out pairs per country (80/20, seed 42)	CO2_RUN.md, manifest.json
Hardware	Single Colab-class GPU (Tesla T4 in the committed run)	DPO_train.ipynb, README.md

D. Protocol artifacts required for a claim-ready version

A claim-ready version will archive: (i) country inclusion rules; (ii) anchor and item-map versions; (iii) prompts, temperatures, model IDs, and endpoint versions; (iv) raw scenario, pair, selection, and scoring outputs; (v) QC tables; (vi) training manifests and adapter hashes; (vii) evaluation outputs; and (viii) analysis code for item, country, generation, and training-seed uncertainty. A remote API seed is not itself a sufficient reproducibility guarantee; materialized datasets are the primary artifacts.

E. Supplementary figures

The figures below are secondary views of the surfaces reported in the main text; they are generated by the same committed pipeline (analysis/phase2/) from the same artifacts.

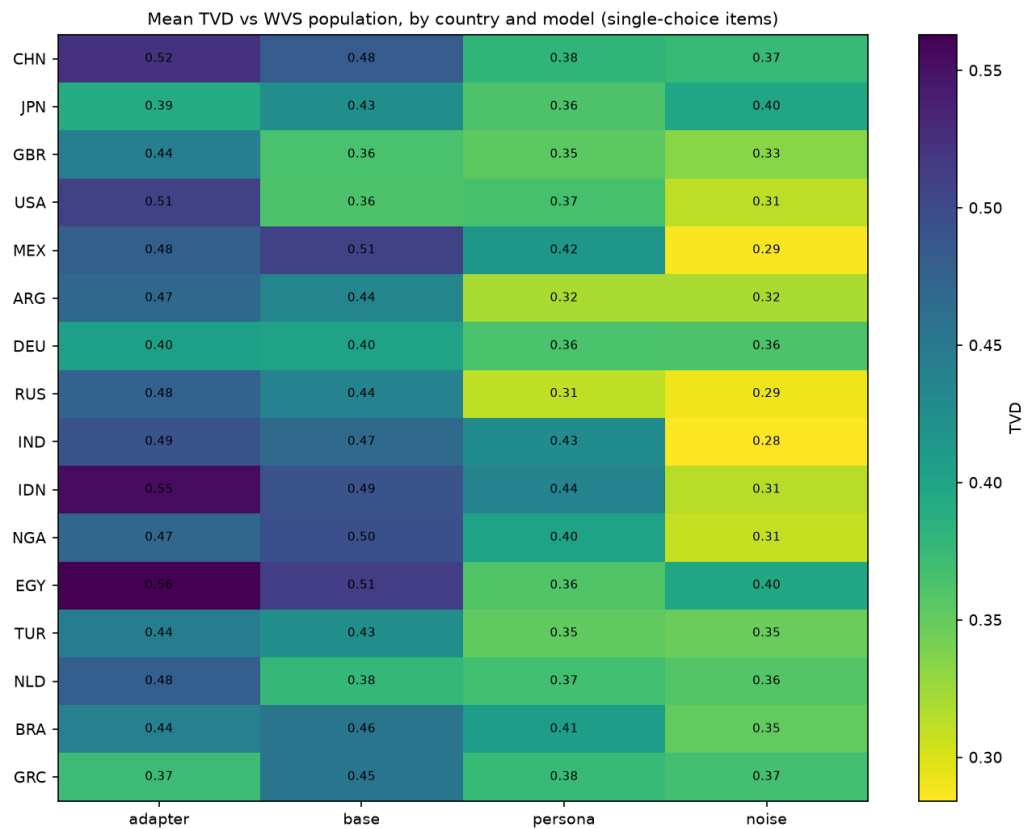


Figure S1: Total variation distance by country and model family, mean over the 30 single-choice items. Darker cells are further from the wvs population; noise is closest on average because the populations are diffuse.

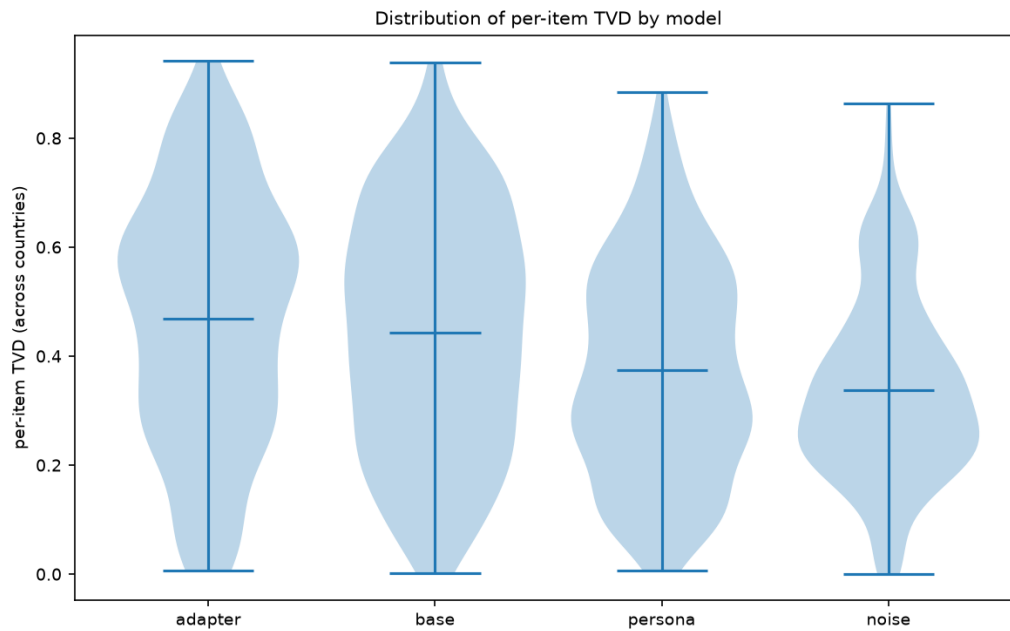


Figure S2: Per-item TVD distributions by model family. The adapters show the widest spread with a high median (over-concentration across items), persona is narrower, and noise has the lowest median and tightest spread.

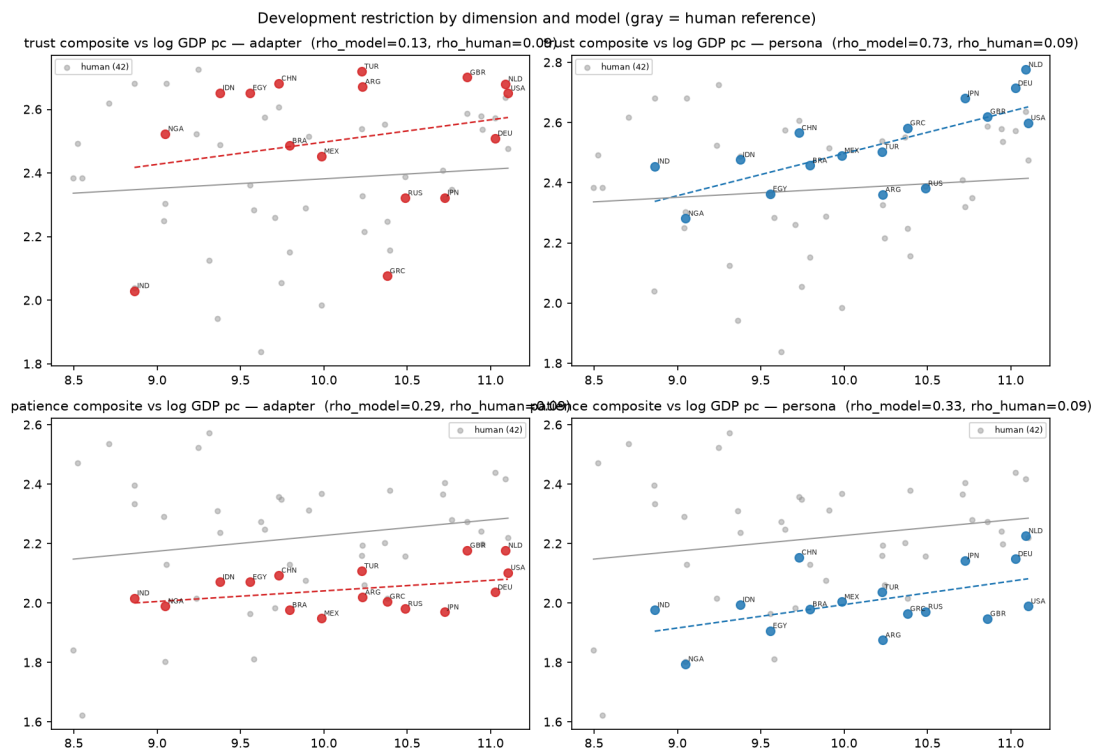


Figure S3: Development restriction: country composite versus log GDP per capita by dimension. Red points are adapters, blue points persona prompts, gray points the 42-country human overlay; Spearman ρ is reported in each panel title.

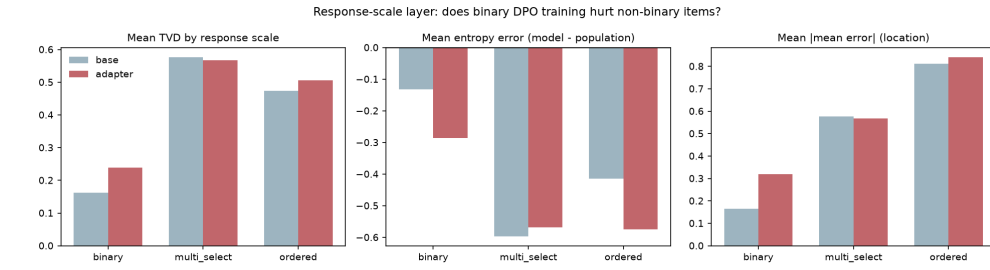


Figure S4: Inference-time temperature scaling: pooled TVD by temperature. Shape (dispersion) is recovered at $T = 3$ while top-option match is unchanged; location is unaffected. Reported as a diagnostic, not a training change.

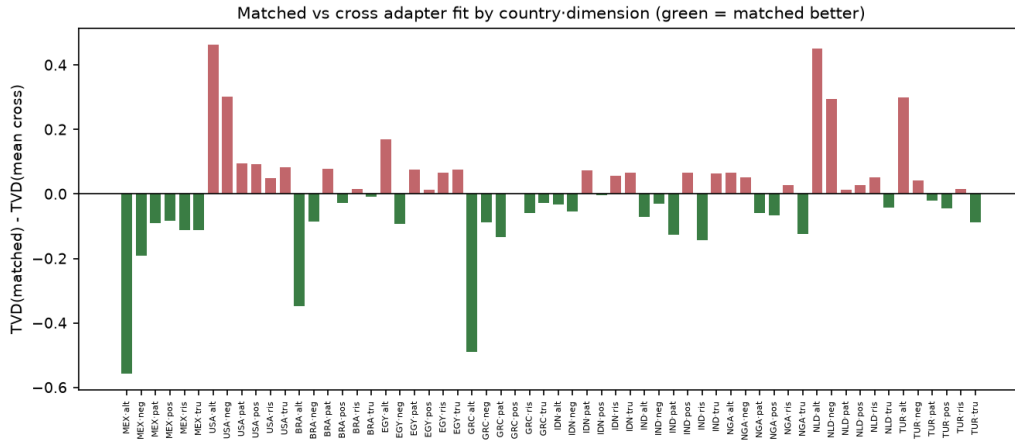


Figure S5: Matched-versus-cross TVD comparisons underlying the two-country pilot (Appendix A).

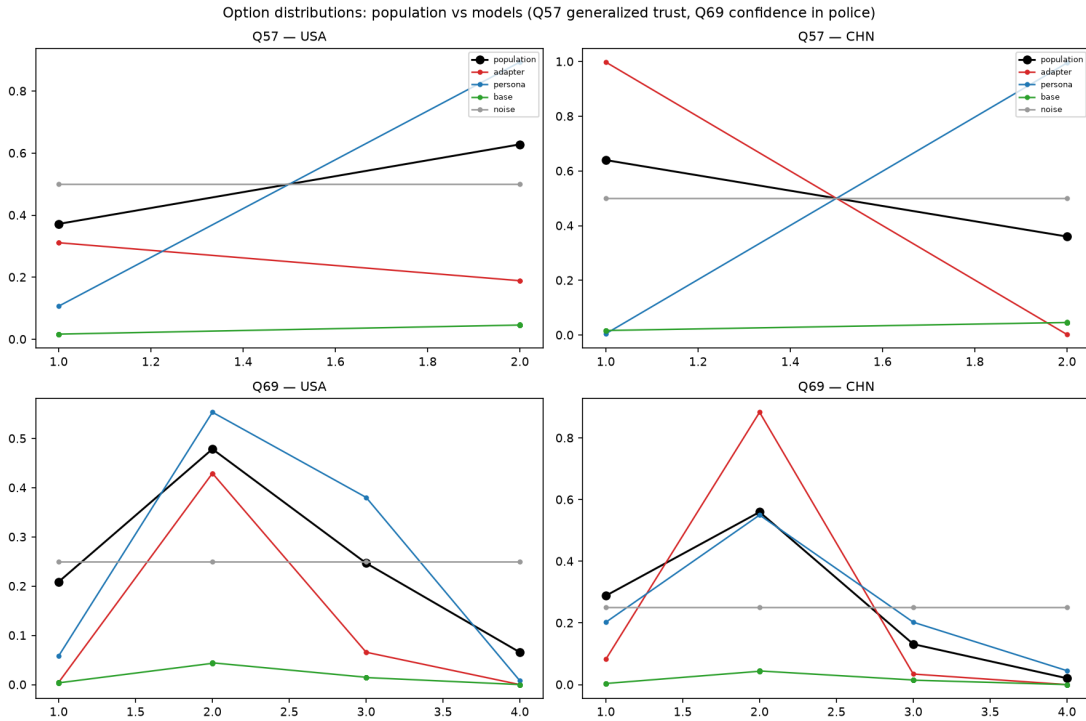


Figure S6: Item option distributions for two illustrative items (Q57 generalized trust; Q69 confidence in police). Black is the wvs population, red the adapter, blue the persona prompt, green the base, gray uniform noise.

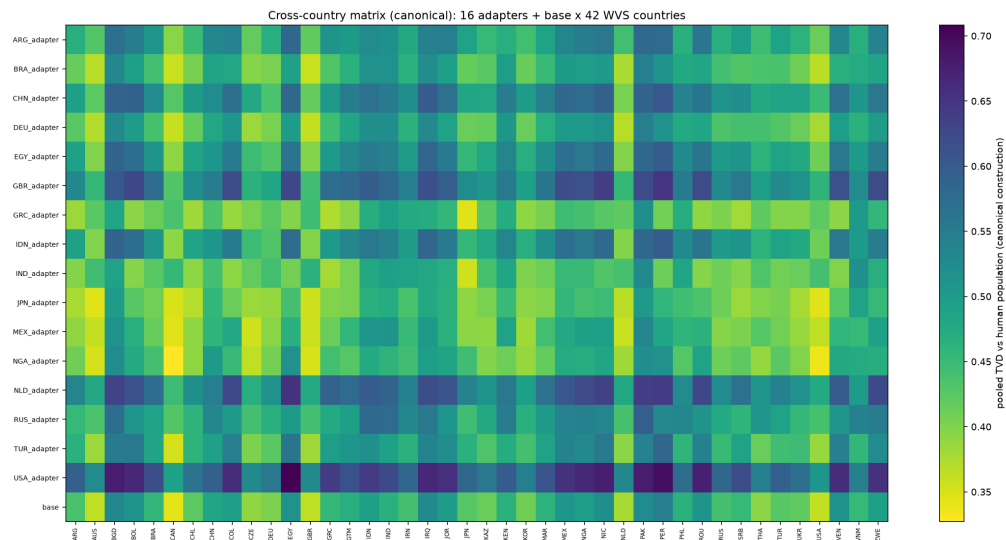


Figure S7: Full cross-country matrix: pooled TVD of each adapter and the base against all 42 WVS populations (canonical construction; own-country cells equal Table 1). Darker cells are further from the population. The row-wise minima are dominated by mid-scale populations, and no adapter is closest to its own country.

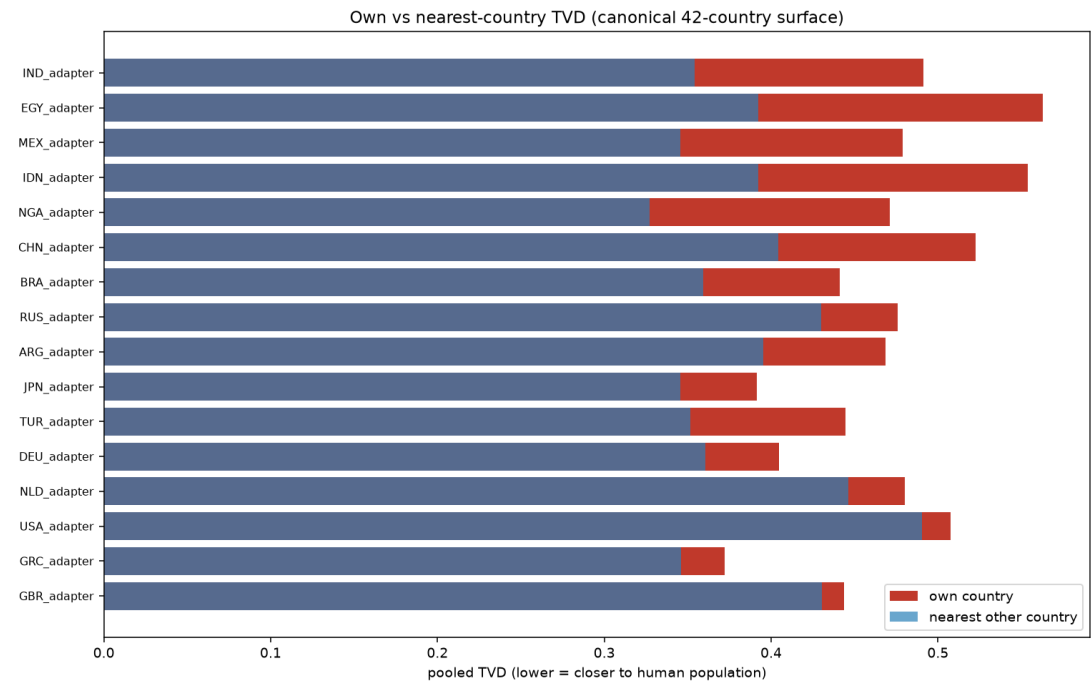


Figure S8: Own-country TVD versus the nearest other country for each adapter. Own-country bars (red) exceed the nearest-other bars (blue) for every adapter; the anti-specific set (NGA, IDN, MEX, EGY, IND) shows the largest gaps.

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